

## EFY Note

The source code of this project is available for free download at [source.efymag.com](http://source.efymag.com)

## Construction and testing

An actual-size PCB layout for the transmitter is shown in Fig. 9 and its components layout in Fig. 10. After assembling the circuit on PCB, connect 5V supply across CON1.

An actual-size PCB layout for receiver is shown in Fig. 11 and its components layout in Fig. 12. After assembling the circuit on PCB, connect another 5V supply across CON3. Use a 5V, 10A contact-rating relay for RL1. As relay RL1 is not PCB-mounted, connect its coil to the PCB where it is marked RL1. Connect the water-pump's power on/off switch across Pole and NO contacts of the relay.

Switch on the transmitter and

## PARTS LIST

|                                              |                                            |
|----------------------------------------------|--------------------------------------------|
| <b>Semiconductors:</b>                       |                                            |
| Board1                                       | - NodeMCU module                           |
| SS1                                          | - Soil sensor module                       |
| IC1                                          | - HT12E encoder                            |
| IC2                                          | - HT12D decoder                            |
| D1, D2                                       | - 1N4007 rectifier diode                   |
| T1                                           | - BC547 npn transistor                     |
| T2, T3                                       | - BC317 npn transistor                     |
| LED1-LED4                                    | - 5mm LED                                  |
| <b>Resistors (all 1/4-watt, ±5% carbon):</b> |                                            |
| R1                                           | - 1-mega-ohm                               |
| R2-R4                                        | - 330-ohm                                  |
| R5, R7, R9-R11                               | - 470-ohm                                  |
| R6                                           | - 47-kilo-ohm                              |
| R8                                           | - 4.7-kilo-ohm                             |
| <b>Miscellaneous:</b>                        |                                            |
| J1                                           | - 3-pin connector with shorting jumper     |
| CON1                                         | - 2-pin connector                          |
| CON2                                         | - 4-pin connector                          |
| CON3                                         | - 2-pin terminal connector                 |
| CON4                                         | - 3-pin connector (not shown in PCB)       |
| CON5                                         | - 3-pin connector                          |
| TX1                                          | - 433MHz RF transmitter module             |
| RX1                                          | - 433MHz RF receiver module                |
| RL1                                          | - 5V, 10A single-changeover relay          |
|                                              | - 3V battery /5V DC supply for transmitter |
|                                              | - 5V DC supply for receiver                |
| ANT.1, ANT.2                                 | - 30cm hook-up wire for antenna            |

## « Do-It-Yourself

receiver circuits. Login to <https://io.adafruit.com> and open the dashboard you have already created. Depending on the soil moisture level in the field, you will see the relay status on the dashboard, as shown in Figs 6 and 7. When Relay1 status is 1, motor is on and when Relay1 status is 0, motor is off. The status of the relay will change automatically depending on the soil moisture level in the field.

For testing/checking the signals in transmitter and receiver circuits (Fig. 2 and Fig. 3), the expected signals at different points are listed in Table 1. **EFY**



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